

1. Introduction

In this module, I explored the foundational elements that shape how modern networks operate. From understanding the structure of the Internet to dissecting the OSI and TCP/IP models, I gained insights into the essential components that allow communication and data exchange to happen efficiently and securely. The module also introduced me to critical networking concepts such as proxies, topologies, and IP addressing, which are fundamental to any cybersecurity practice.

2. Module Sections

Q Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

Answer 255.255.255.224

To answer this question, I first look at the CIDR notation: 10.200.20.0/27. The /27 tells me that 27 bits out of 32 are used for the network portion of the IP address. That leaves 5 bits for the host portion. I know that a subnet mask is made of 1s for the network bits and 0s for the host bits. So for /27, I write: 11111111.11111111.11111111.11100000. I break this into 4 octets and got the answer:

11111111 = 255

11111111 = 255

11111111 = 255

11100000 = 224

Q Submit the broadcast address of the following CIDR: 10.200.20.0/27

Answer 10.200.20.31

To find the broadcast address of the CIDR 10.200.20.0/27, I follow these steps using what I've learned in networking and subnetting.

Since the CIDR is /27, that means 27 bits are used for the network, leaving 5 bits for hosts.

So the number of IP addresses in this subnet is:

$2^5 = 32$ IP addresses

This includes:

1 network address (first IP)

1 broadcast address (last IP)

30 usable host addresses

The starting IP is the network address: 10.200.20.0. Since the subnet includes 32 addresses, I count up:

10.200.20.0 to 10.200.20.31

The last IP in the range is always the broadcast address.

Q Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

Answer 10.200.20.16

To find the network address of the 3rd subnet when I split 10.200.20.0/27 into 4 subnets, here's how I think through it: The original CIDR is /27, which gives:

Total IPs:

$$2^{(32-27)}=32$$

So the whole range goes from: 10.200.20.0 to 10.200.20.31

Now I need to split this into 4 equal subnets.

To split into 4 subnets, I ask:

How many more bits do I need to borrow?

4 subnets = 2^2 so I borrow 2 bits from the host portion

New prefix: $/27 + 2 = /29$

Each /29 subnet has

$$2^{(32-29)}= 8 \text{ IP addresses}$$

I increment by 8 for each subnet and found the 3rd subnet

1st subnet: 10.200.20.0/29

2nd subnet: 10.200.20.8/29

3rd subnet: 10.200.20.16/29

4th subnet: 10.200.20.24/29

Q Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

Answer 10.200.20.15

To find the broadcast address of the 2nd subnet when splitting 10.200.20.0/27 into 4 subnets, I work through it like this

The CIDR is /27, which gives me:

$$2^{(32-27)}= 32 \text{ total IP addresses}$$

I want 4 equal subnets, so I divide 32 by 4 → each subnet gets 8 IP addresses

To do this, I borrow 2 bits from the host portion, making the new prefix /29

Each /29 subnet has 8 IPs, so I count up in blocks of 8 and got the answer

1st subnet:

Network: 10.200.20.0/29

Broadcast: 10.200.20.7

2nd subnet:

Network: 10.200.20.8/29

Broadcast: 10.200.20.15

3rd subnet:

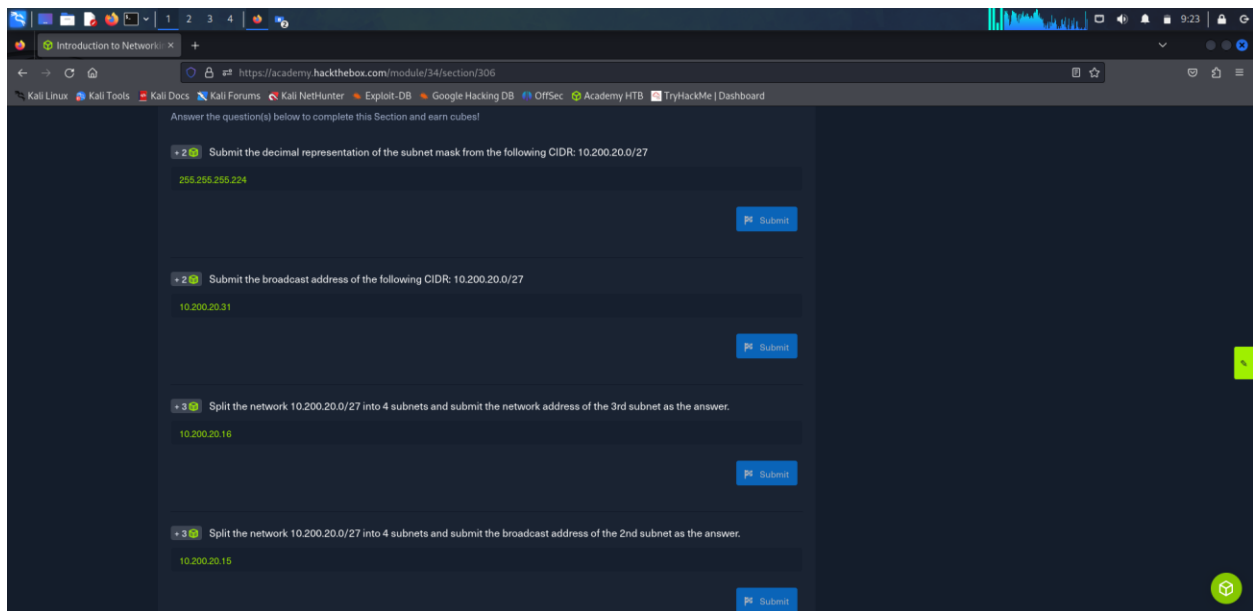
Network: 10.200.20.16/29

Broadcast: 10.200.20.23

4th subnet:

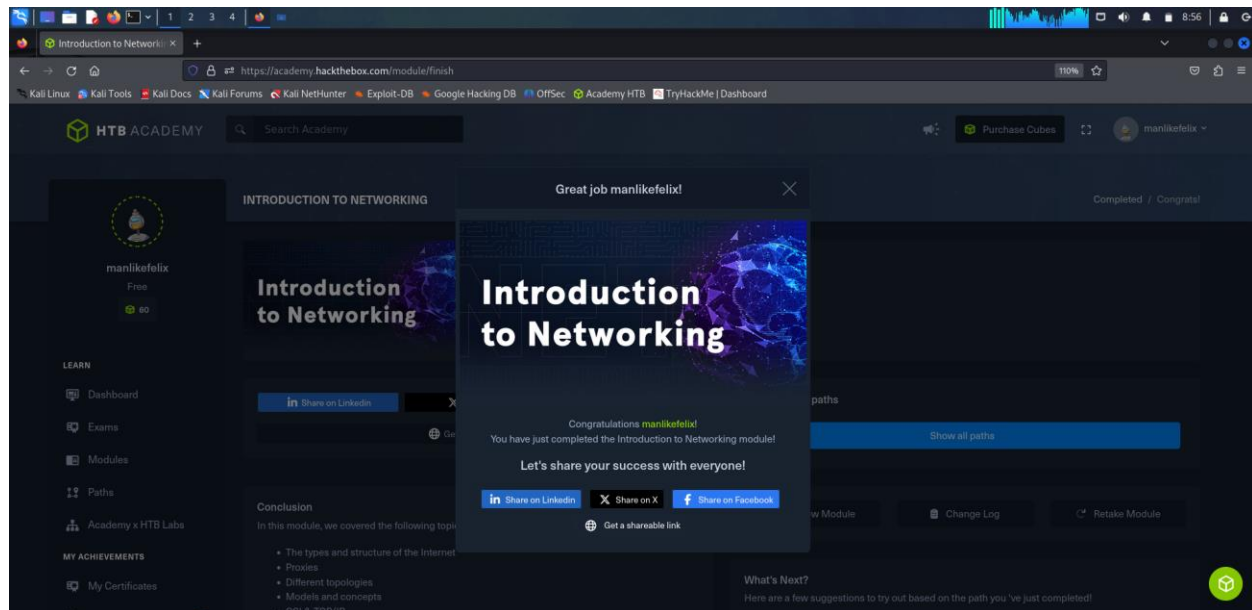
Network: 10.200.20.24/29

Broadcast: 10.200.20.31



3. Shareable Link and Final Screenshot

<https://academy.hackthebox.com/achievement/1919044/34>



4. Social Media Posting

https://www.linkedin.com/uas/login?session_redirect=https%3A%2F%2Fwww.linkedin.com%2FshareArticle%3Furl%3Dhttps%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F34%26title%3DCompleted%2520Introduction%2520to%2520Networking!%26summary%3DI%2520just%2520completed%2520the%2520module%2520Introduction%2520to%2520Networking%2520in%2520HTB%2520Academy!%23hackthebox%23htbacademy%23cybersecurity&source=HTBAcademy

[https://x.com/intent/tweet?url=https://academy.hackthebox.com/achievement/1919044/34&text=I just completed module Introduction to Networking in HTB Academy!&hashtags=hackthebox,htbacademy,cybersecurity](https://x.com/intent/tweet?url=https://academy.hackthebox.com/achievement/1919044/34&text=I%20just%20completed%20module%20Introduction%20to%20Networking%20in%20HTB%20Academy!%23hackthebox%23htbacademy%23cybersecurity)

<https://www.facebook.com/sharer/sharer.php?u=https://academy.hackthebox.com/achievement/1919044/34>

5. Conclusion

This module deepened my understanding of how networks function and interconnect, which is crucial in the field of cybersecurity. By learning about the structure of the Internet, various network topologies, and the role of proxies, I now see how data flows and how security can be applied at different layers. The comparison between the OSI and TCP/IP models clarified how protocols operate and interact. Additionally, studying both IPv4 and IPv6 gave me a broader perspective on addressing, while subnetting challenged me to think critically and apply logic to real-world scenarios. Overall, this module provided a strong technical foundation that I can build upon as I progress in my cybersecurity studies.